

Table A

Cox model of iCCA patient survival using cancer-gene subpaths

Predictor	Number affected ^a	Hazard ratio	95% CI	P value ^b
Presence of metastatic disease	184	2.25	1.81, 2.78	< 0.001***
Subpath mutation in...				
P14★ (TP53, CTNNB1, EGFR...)	172	1.91	1.52, 2.39	< 0.001***
P1★ (KRAS, NRAS, RAC1)	98	1.68	1.30, 2.17	< 0.001***
P25★ (BAP1, PBRM1, SETD2...)	141	0.79	0.62, 1.00	0.053
P33★ (ATR, PTEN, RPTOR...)	73	0.67	0.48, 0.94	0.020*

^a $n = 636$ patients (439 events)

^b * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$

Table B

LASSO-regularized Cox model of iCCA patient survival using whole-exome and cancer-gene subpaths

Predictor	Number affected ^a	Hazard ratio
Presence of metastatic disease	11	1.51
Supbath mutation in...		
P15 (FBN1, FN1, ITGB8...)	14	1.85
P1★ (KRAS, NRAS, RAC1)	31	1.48
P38★ (ARID2, ARID1B, SMARCA4...)	10	1.32
P4★ (SMAD4, BRAF, TGFBR1...)	20	1.3
P27★ (ERBB3, ERBB2, PIK3R1...)	14	1.22
P21 (SPTBN1, SOS1, ITCH...)	14	1.19
P14★ (TP53, CTNNB1, EGFR...)	57	1.18
P2★ (PIK3CA, MTOR, AKT1...)	8	1.17
P8 (MCM8, PSMD3, PSMA5...)	8	0.82
P34★ (APC, CDKN2A, TP63...)	8	0.7
^a n = 171 patients (111 events)		

Table C

Cox model of pan-CCA patient survival using cancer-gene subpaths

Predictor	Number affected ^a	Hazard ratio	95% CI	P value ^b
Presence of metastatic disease	200	2.35	1.92, 2.87	< 0.001***
Subpath mutation in...				
P14★ (TP53, CTNNB1, EGFR...)	211	1.74	1.41, 2.14	< 0.001***
P1★ (KRAS, NRAS, RAC1)	107	1.41	1.11, 1.79	0.005**
P33★ (ATR, PTEN, RPTOR...)	87	0.76	0.56, 1.03	0.075
P25★ (BAP1, PBRM1, SETD2...)	152	0.73	0.58, 0.93	0.009**

^a $n = 722$ patients (502 events)

^b * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$

Table D

LASSO-regularized Cox model of pan-CCA patient survival using whole-exome and cancer-gene subpaths

Predictor	Number affected ^a	Hazard ratio
Presence of metastatic disease	16	1.37
Supbath mutation in...		
P15 (FBN1, FN1, ITGB8...)	14	1.17
P14★ (TP53, CTNNB1, EGFR...)	67	1.14
P1★ (KRAS, NRAS, RAC1)	34	1.04
^a n = 190 patients (126 events)		